

PIG: GPU Redesign of a Plenoptic Imaging Pipeline and Extension to Multi-View Reconstruction

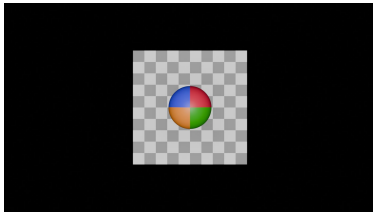
Niccolò Benedetto

Computing Project 2025–2026
Université Libre de Bruxelles

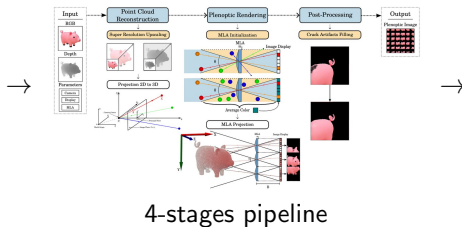
June 2026

Context and Existing Prototype

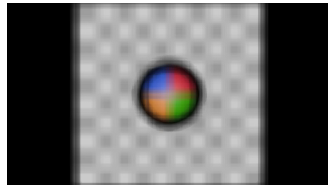
RGB + Depth



PIG Pipeline

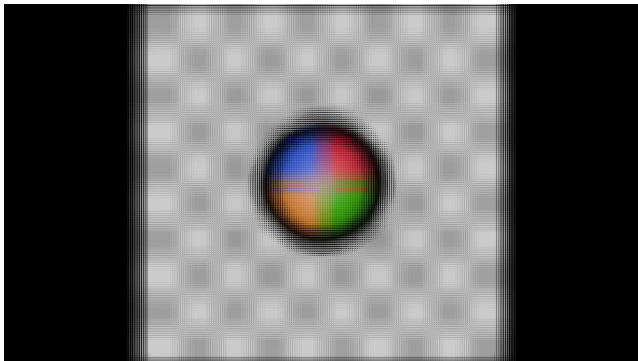


Plenoptic Image



- **Integral Imaging:** glasses-free 3D visualization based on light-field reconstruction through a Microlens Array (MLA)
- **Plenoptic Image Generator (PIG):** converts RGB + Depth data into a plenoptic image, where each microimage is projected by a microlens from a slightly different viewpoint, enabling depth perception and motion parallax

Limitations of the Existing Pipeline



Performance for ball dataset

- CPU execution: ≈ 1600 -1700 ms
- Sequential processing

Visual Artifacts

- Crack artifacts caused by discrete nature of the point-cloud
- Disocclusions near sphere boundary due to missing geometry

Contribution 1

GPU Redesign

- Point Cloud Generation
- Post-Processing
- CUDA optimization
- Nsight profiling

Contribution 2

Multi-View Reconstruction

- Multi-camera RGB+D data support
- New GPU pipeline stages
- Disocclusion recovery

GPU Pipeline Redesign Strategy

Stage	CPU(ms)
Pre-Processing	65.2
Point Cloud Gen	255.8
Plenoptic Rendering	548.3
Post-Processing	844.0
Total	1713.4

CPU Bottlenecks: Point Cloud Generation + Post-Processing

Point Cloud Generation

- Validity Mask Kernel
- CUB Inclusive Scan
- Projection and Scatter Kernel

Post-Processing

- Crack Filtering Kernel
- ROI-based execution
- Rotation Kernel

Key Idea

Independent pixel/point operations → CUDA parallelism

$\approx 5.8\times$

Overall Speedup

Rendering Consistency

PSNR (Post-Processing)

33.9 dB

PSNR (Point Cloud Generation)

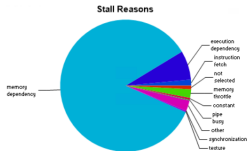
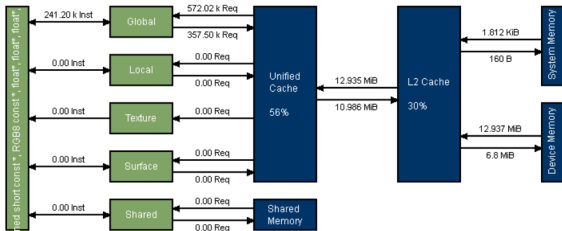
31 dB

GPU output remains visually consistent with CPU baseline

Stage	CPU(ms)	GPU(ms)
Pre-Processing	65.2	26.8
Point Cloud Gen	255.8	128.1
Plenoptic Rendering	548.3	36.5
Post-Processing	844.0	101.7
Total	1713.4	293.0

Project/Scatter Kernel Profiling

Variable	Achieved	Theoretical	Device Limit	Grid Size: [8100,1,1] (8100 block)(Block Size: [256,1,1] (256 thread))
Occupancy Per SM				
Active Blocks		8	32	
Active Warps	52.65	64	64	
Active Threads		2048	2048	
Occupancy	82.3%	100%	100%	



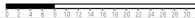
Observations

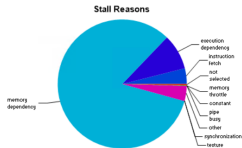
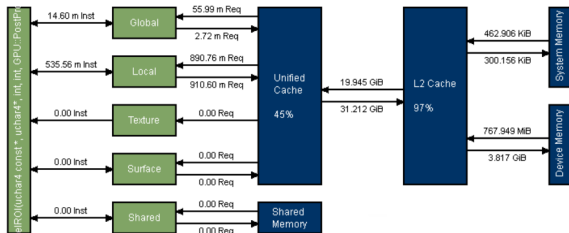
- Occupancy: **82.3%**
- High thread utilization
- Dominant stall reason: memory dependency
- Global memory transactions dominate execution

Conclusion

Kernel is memory-bound rather than compute-bound.

Crack Filtering Kernel Profiling

Variable	Achieved	Theoretical	Device Limit	Grid Size: [1,1,65535] (65535 blocks)(Block Size: [16,16,1] (256 threads))
Occupancy Per SM				
Active Blocks		8	32	
Active Warps	56.7	64	64	
Active Threads		2048	2048	
Occupancy	88.6%	100%	100%	



Observations

- Occupancy: **88.6%**
- L2 Cache utilization: **97%**
- Memory dependency dominates warp stalls
- Large cache-device memory traffic

Conclusion

Kernel approaches memory bandwidth saturation while maintaining high occupancy.

GPU Multi-View Reconstruction

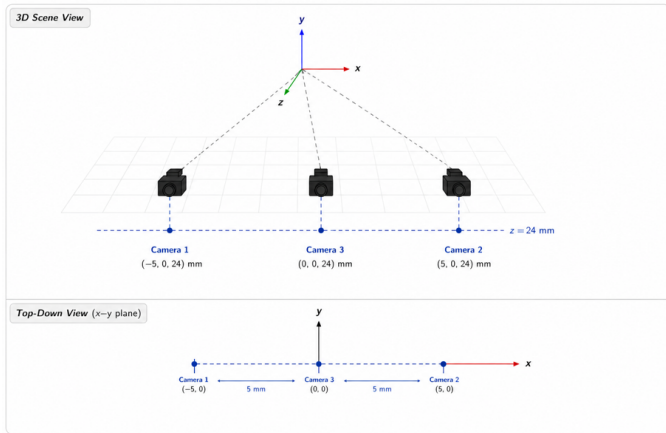
New Pipeline

- 1 Point Cloud Generation
- 2 Registration and Merging
- 3 Plenoptic Rendering
- 4 Post Processing

Rigid Transformation (stage 2)

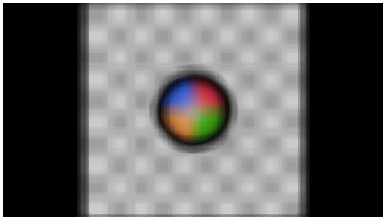
$$p_{world} = R^{-1}p_{camera} + C$$

One CUDA thread processes one point independently.

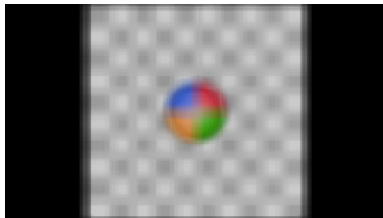


Multi-View Qualitative Results

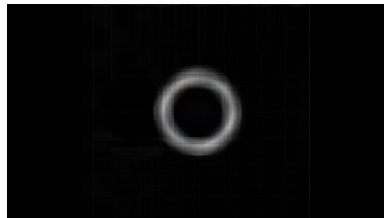
Single-View



Multi-View



Difference



- Dark boundaries around the sphere significantly reduced
- These regions correspond to disocclusions caused by missing geometry

350,023 Recovered Pixels

$$\text{Recovery Ratio} = \frac{\text{RecoveredPixels}}{\text{InvalidPixelSingleView}} = 5.16\%$$

53.98% $\rightarrow$ 56.23%

$$\text{Valid Pixel Density} = \frac{\text{InvalidPixels}}{\text{TotalPixels}}$$

Registration Kernel

- Occupancy: 86.7%
- High thread utilization
- Memory dependency
- Low arithmetic cost

Variable	Achieved	Theoretical	Device Limit	Grid Size: [1313,1,1] (1313 blocks)Block Size: [256,1,1]
Occupancy Per SM				
Active Blocks		8	32	
Active Warps	55.49	64	64	
Active Threads		2048	2048	
Occupancy	86.7%	100%	100%	
Warps				
Threads/Block		256	1024	
Warps/Block		8	32	
Block Limit		8	32	
Registers				
Registers/Thread		26	255	
Registers/Block		8192	65536	
Block Limit		8	32	
Shared Memory				
Shared Memory/Block		0	96304	
Block Limit		0	32	

$\approx 5.8\times$

GPU Speedup

≈ 34 dB

Rendering Correctness

$\approx 350k$

Recovered Pixels

GPU acceleration of PIG & Multi-View Geometry Recovery

Future Work: memory optimization, larger datasets